

Lab 3: Architectural Pattern Selection — Written Justification

Group Travel Expense Simplification Engine

Architecture Selection: We chose Layered Architecture for the Group Travel Expense Simplification Engine System.

Architectural Choice

The system is organised into three horizontal layers: a Presentation Layer (User Interface Component) for creating trip groups and recording expenses; a Business Layer (Expense Manager, Currency Converter, Settlement Engine, and Payment Confirmation Components) that implements the core epics — expense recording & splitting, multi-currency conversion, minimum settlement calculation, and settlement/payment confirmation; and a Data Layer (Database Component) that persists trip groups, members, expenses, exchange rates, and settlement records.

Reasons for the Choice

- **A predictable, sequential pipeline:** Expense entry, currency normalisation, settlement calculation, and payment confirmation naturally form a pipeline where each step builds on data the previous one produced. Grouping these as business-layer components keeps that pipeline coherent and easy to test end-to-end, without the deployment and network overhead microservices would add for what is a single-team, single-trip workflow.
- **Shared, consistent trip data:** Expense recording, currency conversion, settlement calculation, and payment confirmation all operate on the same trip group's data. A layered design lets all four business components reach one consistent Database Component, avoiding the data-consistency problems of splitting this tightly coupled data across separate microservice databases — a settlement computed from stale expense data would produce wrong payment amounts.

Security Advantage

Because the Presentation Layer can only reach the Database Component indirectly, through the ExpenseInterface, ConversionInterface, and SettlementInterface, the Business Layer becomes the single point that verifies a traveler belongs to a trip group before letting them add, convert, or settle an expense. The UI never queries the database directly, so a compromised client cannot bypass this check to read or alter another traveler's expenses or settlement records.

Performance Benefit

Daily exchange rates change at most once a day, and a settlement graph only needs recalculating when a trip's expenses change. The Currency Converter and Settlement Engine Components can cache these results within the Business Layer, avoiding repeated rate lookups or full settlement recalculations every time a traveler simply views the settlement summary — keeping the app responsive even for trip groups with many expenses.